

Animal Adoption Analysis

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Summary

Animal adoption is a critical issue for shelters, as many animals experience prolonged stays or never find a permanent home. This study aims to build a classification model that predicts whether an animal will be adopted based on its attributes. Using data from the [Long Beach Animal Shelter](#) (TidyTuesday Project (2025)), we will analyze key factors such as species, breed, age, health status, and intake type. Through exploratory data analysis and predictive modeling, we hope to identify patterns that influence adoption rates. Our goal is to provide insights that can help shelters improve adoption strategies and increase successful placements.

Introduction

Background Information

Animal shelters play a crucial role in rehoming stray and surrendered animals. However, many animals face extended shelter stays, leading to overcrowding and resource constraints (ASPCA (2025)).

Recent studies indicate that **adoption rates vary significantly by species, breed, and medical condition** (The Zebra (2025)). Understanding what factors influence adoption can help shelters make data-driven decisions to improve adoption rates.

In recent years, data-driven approaches have been applied to predict adoption outcomes, allowing shelters to prioritize resources and create targeted adoption campaigns (BMC Veterinary Research (2020)). Previous studies have shown that characteristics such as species, age, breed, and health status can significantly impact adoption likelihood (DrPress (2021)). By leveraging machine learning techniques, we aim to develop a model that predicts whether an animal will be adopted based on its attributes.

Problem Statement

Can we predict whether an animal will be adopted based on its attributes? This study aims to address this question using a classification model trained on animal shelter intake and outcome data. Specifically, we will classify each animal as either **adopted (Yes)** or **not adopted (No)** using key features such as species, age, breed, and medical conditions.

By understanding which features are most important in predicting adoption, we can provide actionable insights to shelters to increase adoption rates and improve animal welfare.

Dataset Description

The dataset used in this study comes from the **TidyTuesday project** and contains **animal intake and outcome records** from the **Long Beach Animal Shelter**. The dataset provides details on each animal's characteristics, the reason for intake, and the outcome of their stay at the shelter.

This dataset is publicly available and can be accessed here:

[TidyTuesday Project \(2025\)](#).

The original dataset consists of 29,787 records with 22 variables describing animal shelter operations from 2017 to 2024.

Key Variables

Below are some of the key variables from the original dataset that are relevant to predicting adoption outcomes (after removing variables with limited analytical value):

Table 1: Key Variables Used in Analysis

Variable	Description
animal_type	Type of animal (e.g., Dog, Cat)
sex	Altered sex status of the animal (e.g., Male/Neutered, Female/Spayed)
dob	Date of Birth (some entries contain future dates)
intake_condition	Condition at intake (e.g., Healthy, Sick, Injured)
intake_type	Reason for intake (e.g., Stray, Owner Surrender)
outcome_date	Exit or Outcome date (e.g., date of adoption, euthanasia, or transfer)
outcome_type	Original outcome category for the animal (e.g., Adoption, Euthanasia, Transfer, Return to Owner)

- Feature Engineering: Derived Variables

To improve predictive modeling, we created two new variables:

- **age** → Derived from **dob**, representing the animal’s age in years.
- **season** → Extracted from **outcome_date** to analyze adoption trends across seasons.

- Identifying the Target Variable (adopted)

Our project aims to predict whether an animal will be adopted. Since the dataset does not have a direct column for this, we derived our target variable from the **original outcome_type** values as shown below:

Table 2: Distribution of Major Outcome Types

Outcome Type	Count
rescue	6680
adoption	6290
euthanasia	5451
transfer	4869
return to owner	3214
shelter, neuter, return	919
died	763
community cat	386
return to wild habitat	294
transport	206
NA	187
foster to adopt	166

As shown in the table above, the original dataset contains a variety of outcome types. “Rescue” is the most common outcome (6,680 animals), followed closely by “adoption” (6,290 animals). The dataset also contains 187 records with missing outcome types (“NA”), which will be addressed during data preprocessing below.

For our predictive modeling task, we simplified these diverse outcomes into a binary target variable called **adopted**:

- **adopted** = "Yes" if “outcome_type” is “**adoption**” or “**foster to adopt**”
- **adopted** = "No" for all other outcome types (including “rescue”, “euthanasia”, “transfer”, etc.)

This transformation allows us to focus on the specific question of whether an animal will be adopted, rather than predicting all possible outcomes. And the class imbalance in our target variable (with more “No” than “Yes” cases) will be addressed during the data preprocessing and modeling phases.

Methods & Results

Our analysis follows a systematic workflow starting with data preprocessing, followed by exploratory data analysis, and feature engineering based on insights discovered. Each step informed subsequent decisions to prepare the data for predictive modeling.

Initial Data Preprocessing

To prepare our raw data for analysis, we performed several essential preprocessing steps:

1. **Handling Missing Values:** We removed records with missing outcome types (187 records) as these were critical for constructing our target variable. Similarly, records with missing dates of birth (3591 records) were excluded since age calculation would be impossible without this information.
2. **Removing Invalid Dates:** We found that dates of birth in the original dataset ranged from 1993-05-28 to 2031-03-30, with some dates extending far into the future. We filtered out records with impossible future dates of birth (beyond December 31, 2024) as these clearly represented data entry errors that would skew our age calculations.
3. **Feature Creation:** We calculated each animal’s age in years based on their date of birth, providing a more interpretable metric than raw dates:
 - **Age** was derived by calculating the difference between the current date and birth date, then converting days to years
 - We also extracted **month** information from outcome dates and mapped these to **seasons** (Spring, Summer, Fall, Winter) to capture potential seasonal adoption patterns
4. **Feature Selection:** We dropped columns with limited analytical value or high missing rates:
 - Identifiers (**animal_id**) and names (**animal_name**) were removed as they don’t predict adoption outcomes
 - Geographical data (**latitude**, **longitude**, **jurisdiction**, **crossing**) was excluded as location analysis was outside our study scope
 - Variables with extensive missing data (**secondary_color**, **reason_for_intake**) were removed to maintain data quality

- Redundant or derivative fields (`was_outcome_alive`, `outcome_subtype`) were excluded to simplify the dataset

These preprocessing steps resulted in a cleaned dataset containing 26,010 records with 16 variables.

Exploratory Data Analysis

Following data preprocessing, we conducted exploratory data analysis to understand key patterns and relationships in the data. This exploratory phase guided our subsequent feature engineering decisions.

1. Adoption Rates Overview

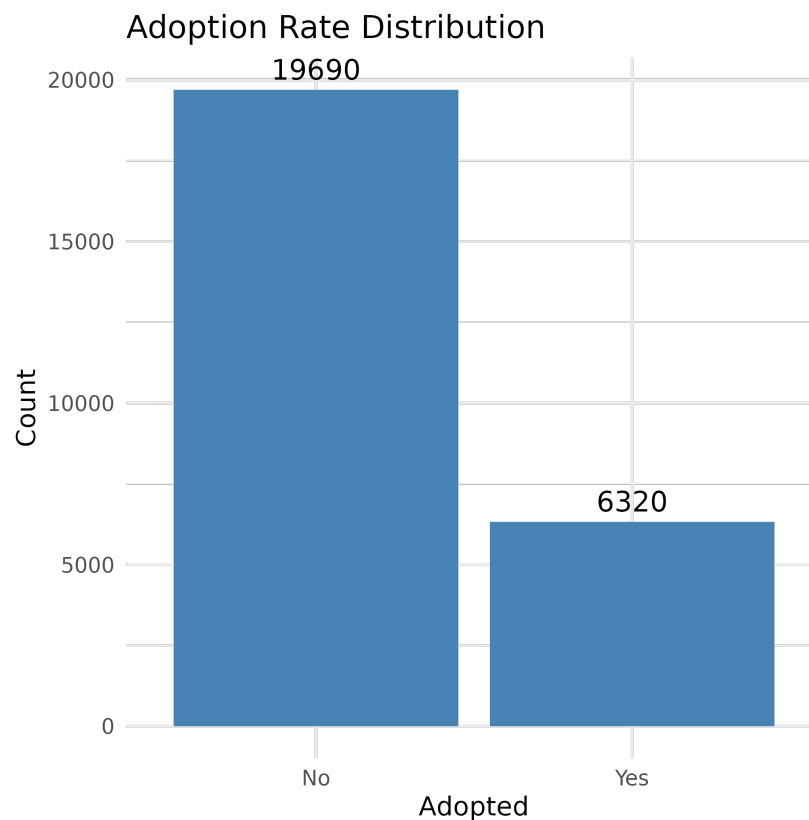


Figure 1: Distribution of adoption outcomes showing class imbalance in the dataset

As shown in Figure 1, there is a significant class imbalance in our target variable, with approximately 75.702% of animals not being adopted. This imbalance informed our modeling approach, requiring techniques to address the disproportionate class representation.

2. Distribution of Animal Types and Adoption Rates

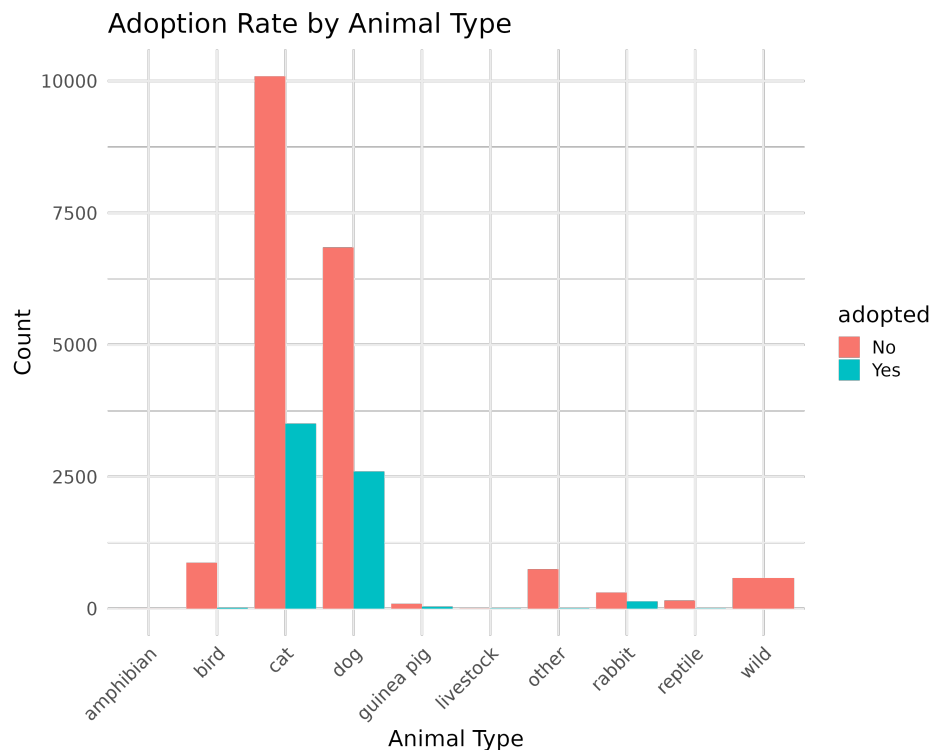


Figure 2: Adoption rates by animal type, showing dogs and cats have the highest adoption counts

Figure 2 shows that dogs and cats form the majority of shelter animals, with dogs having higher adoption rates proportionally. Other animal types like birds, rabbits, and reptiles have significantly lower adoption numbers. This analysis led us to group rare animal types (reptiles, guinea pigs, livestock, and amphibians) into an “Other” category to improve model performance.

3. Intake Characteristics and Adoption: Intake Conditions and Intake Types

The analysis of intake characteristics (Figure 3 and Figure 4) revealed significant patterns. Healthy animals (“normal”) have the highest adoption rates, while animals with medical conditions or behavioral issues show lower adoption rates. Similarly, the manner in which animals enter the shelter affects their adoption outcomes - owner-surrendered animals show higher adoption rates than strays, while animals from wildlife, confiscation, or quarantine cases are rarely adopted. Based on these analyses, we grouped rare intake conditions and types (those with fewer than 200 instances) into an “Other” category during feature engineering.

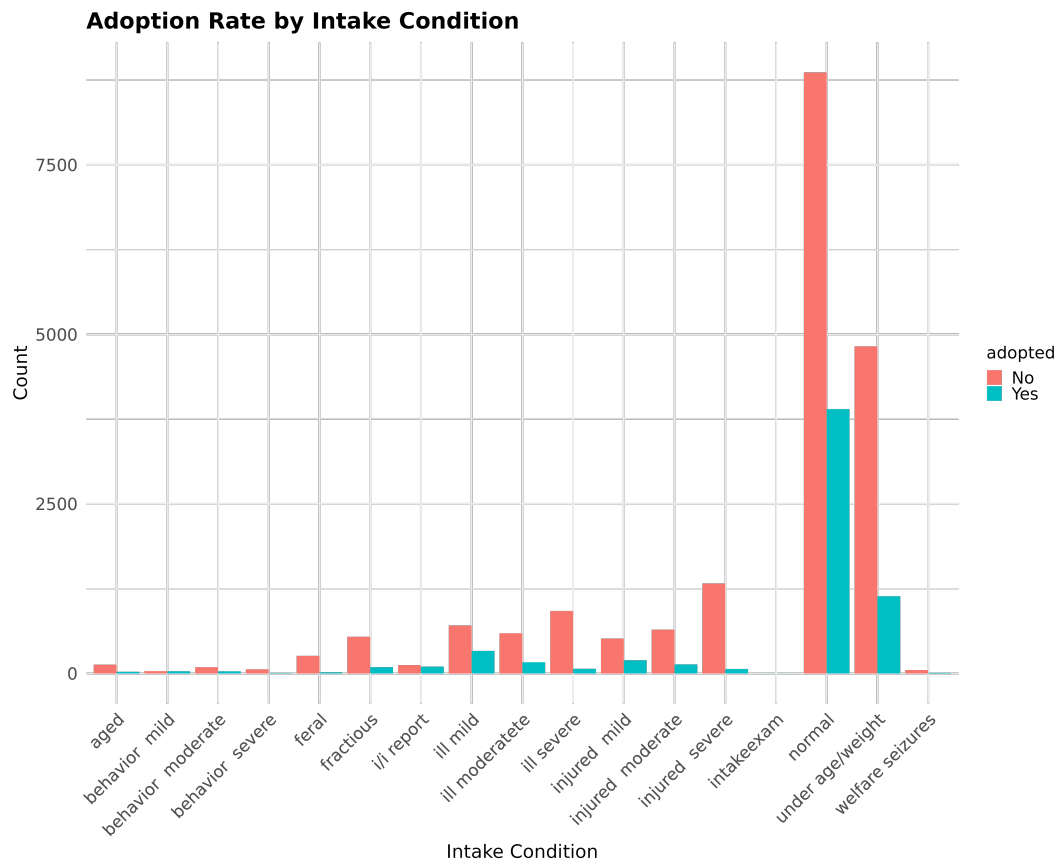


Figure 3: Adoption rates by animal intake condition

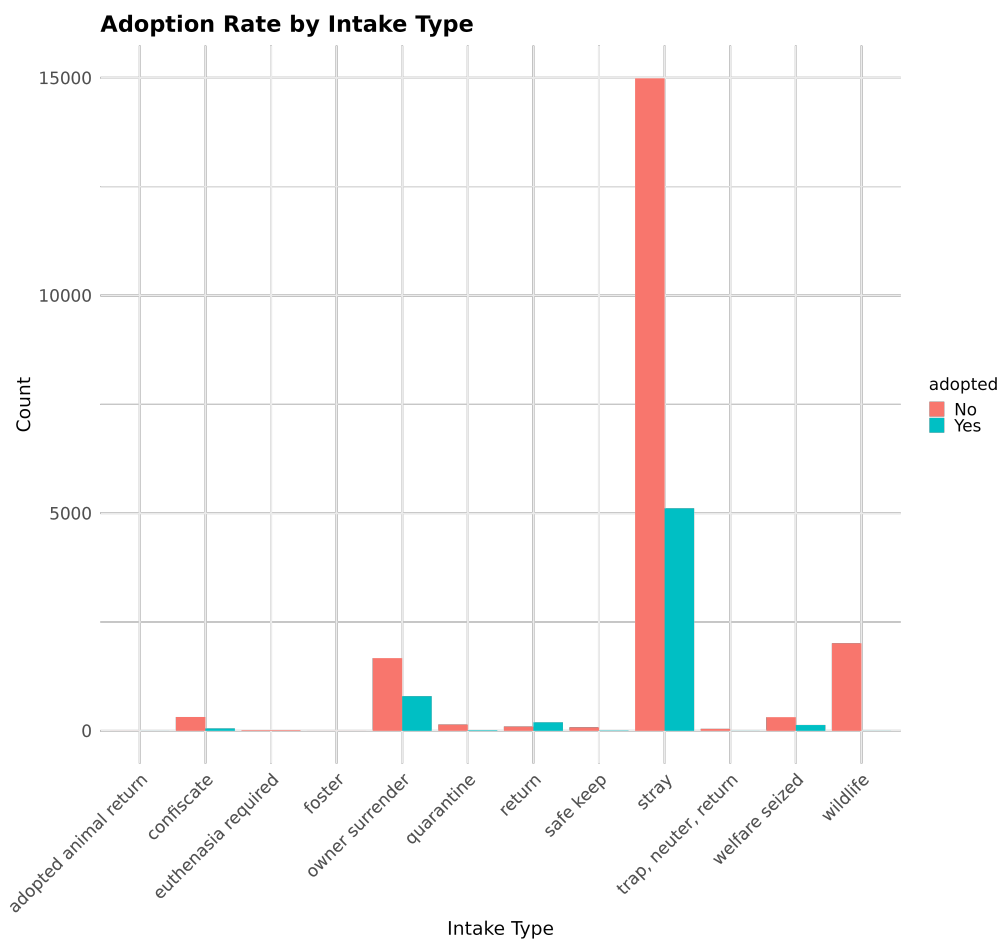


Figure 4: Adoption rates by intake type

4. Age Distribution and Adoption

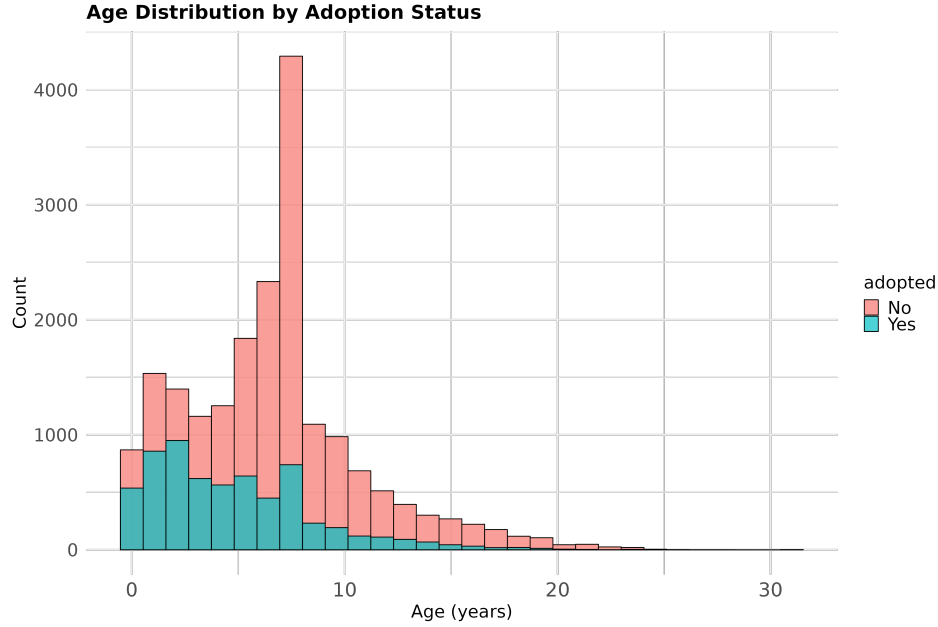


Figure 5: Histogram of age distribution by adoption status

The age distribution histograms (Figure 5) reveal that younger animals represent the majority of the shelter population and have higher adoption rates. The boxplot visualization (Figure 6) further highlights this trend while also revealing outliers with ages exceeding 30 years, which were subsequently removed during data preprocessing to improve model performance.

5. Seasonal Trends

Seasonal analysis (Figure 7) revealed modest variations in adoption rates throughout the year. Summer months showed slightly higher adoption rates, while winter had the lowest rates, suggesting potential for seasonal-targeted adoption campaigns.

Feature Engineering and Data Transformation

Based on insights from our exploratory analysis, we implemented several data transformations to prepare the dataset for modeling:

Handling Rare Categories

Our EDA revealed several categorical variables with rare classes (fewer than 200 instances). To prevent overfitting and improve model generalization, we consolidated these rare categories:

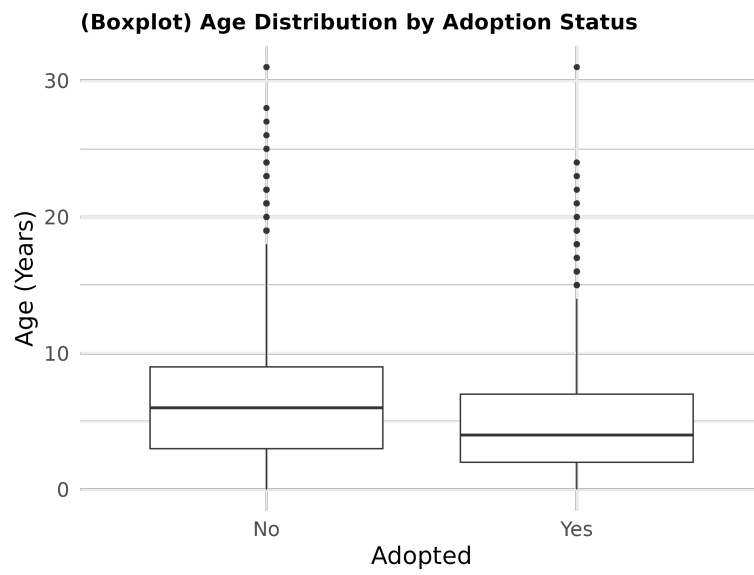


Figure 6: Boxplot showing age distribution by adoption status

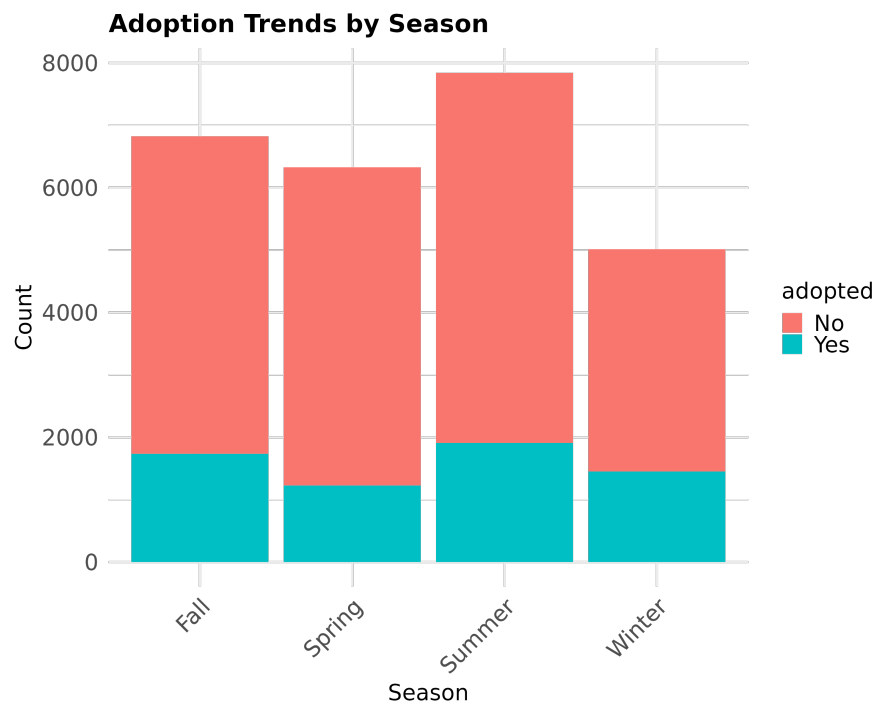


Figure 7: Adoption patterns across different seasons

- **Animal Types:** Consolidated reptiles, guinea pigs, livestock, and amphibians into an “Other” category
- **Intake Conditions:** Combined aged, behavior-related conditions, welfare seizures, and intake exams into an “Other” category
- **Intake Types:** Grouped foster, adopted animal returns, euthanasia required, trap-neuter-return, safe keep, and quarantine into an “Other” category

Outlier Removal

As identified in our age distribution analysis (Figure 6), we removed animals with recorded ages exceeding 30 years, as these likely represented data entry errors that could negatively impact model performance.

Data Type Conversion

All categorical variables were converted to factors to ensure proper handling by our modeling algorithms: `animal_type`, `sex`, `intake_condition`, `intake_type`, `season`, `adopted` (target variable)

These transformations resulted in a clean, structured dataset with 26007 observations and 7 features ready for predictive modeling.

Modeling

To predict animal adoption outcomes, we developed a classification model following these steps:

Random Forest Classifier

We selected a Random Forest model due to its robustness with categorical features and ability to handle non-linear relationships. The model was trained using the following approach:

1. **Train-Test Split:** The dataset was split into 80% training and 20% testing sets using stratified sampling to maintain the same class distribution in both sets.
2. **Class Balance:** As shown in Figure 1, with only 24.298% of animals being adopted. To prevent the model from biasing toward the majority class, we applied downsampling to the training data:
 - The majority class (non-adopted animals) was randomly downsampled to match the number of adopted animals.
 - This resulted in a balanced training dataset with equal representation of both classes.
 - The test set was kept in its original distribution to reflect real-world conditions.

- This approach ensures the model learns equally from both classes while being evaluated on a realistic data distribution.
3. **Model Parameters:** We used 100 trees (`ntree=100`) and 2 variables sampled at each split (`mtry=2`)

Model Performance Evaluation

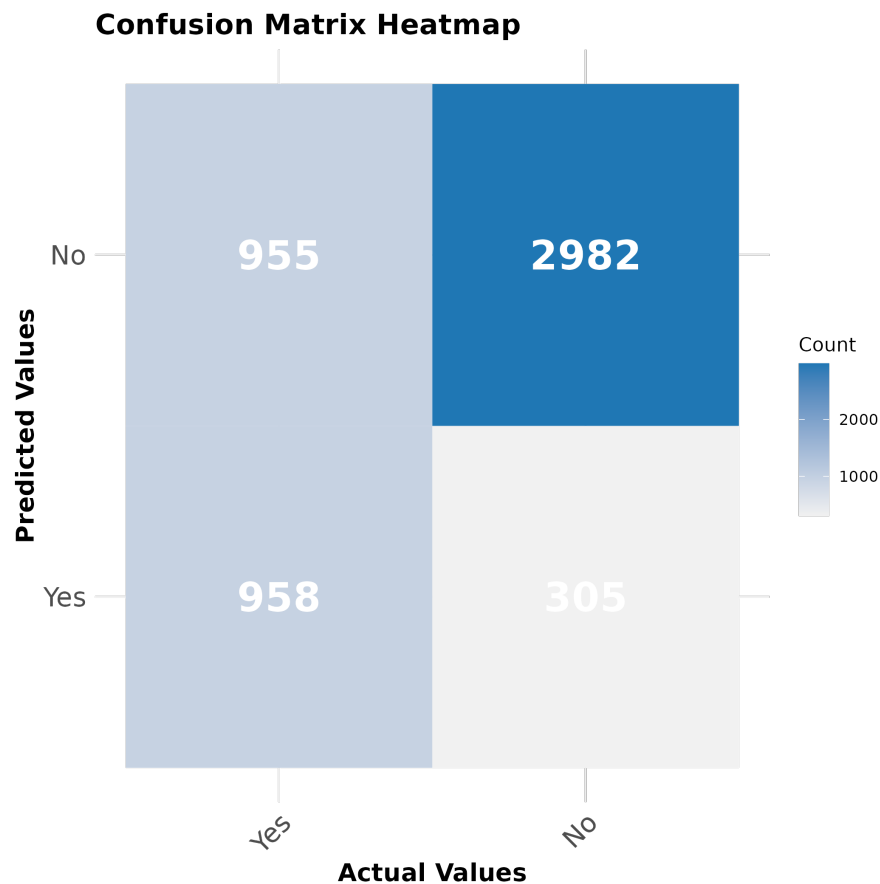


Figure 8: Confusion matrix showing prediction results for animal adoption

The confusion matrix (Figure 8) visualizes our model's performance on the test dataset. Our Random Forest classifier achieved an overall accuracy of 75.8%. The model demonstrates a sensitivity (true positive rate) of 75.9%, indicating its ability to correctly identify animals that were actually adopted. The specificity (true negative rate) of 75.7% shows the model's effectiveness in identifying animals that were not adopted.

Looking at the raw counts in the confusion matrix, we can see that out of animals actually adopted (Yes), the model correctly identified 958 cases while misclassifying 955 as not adopted.

For animals that were not adopted (No), the model correctly identified 2982 cases while misclassifying 305 as adopted. These results suggest the model performs slightly better at identifying non-adopted animals, which aligns with the class distribution in our original dataset.

Feature Importance

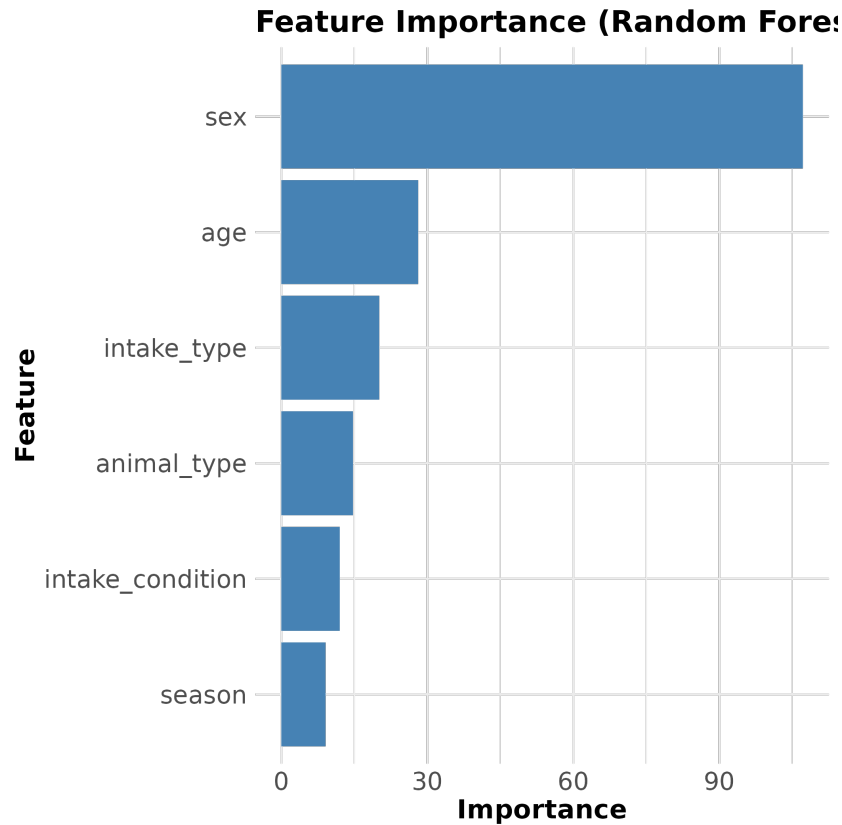


Figure 9: Feature importance for the Random Forest model

Figure 9 reveals the relative importance of each feature in predicting adoption outcomes. The most influential factors in determining whether an animal will be adopted are:

1. Sex - The animal's sex and reproductive status (neutered/spayed) emerged as the most important predictor.
2. Age - Younger animals show significantly higher adoption rates.
3. Intake type - How an animal arrived at the shelter (stray, owner surrender, etc.) substantially affects adoption outcomes.
4. Animal type - Different species have varying adoption rates.
5. Season - Time of year shows a modest effect on adoption probability.

6. Intake condition - The animal's health status when entering the shelter influences adoption likelihood.

This feature ranking provides valuable insights for shelter management, suggesting that factors like **sex** and **age** are particularly important considerations when developing targeted adoption strategies.

Discussion

Our analysis of the Long Beach Animal Shelter data revealed that sex/reproductive status, age, and intake type are the most influential factors in animal adoption outcomes. The strong influence of sex suggests that spay/neuter status significantly impacts adoption decisions, while the consistent preference for younger animals presents an ongoing challenge for shelters placing older pets. Additionally, the manner in which animals enter the shelter affects their adoption prospects, with owner-surrendered animals generally finding homes more easily than strays or confiscated animals.

Conclusion

This study successfully developed a predictive model for animal adoption outcomes at the Long Beach Animal Shelter. Our Random Forest classifier identified key factors influencing adoption, with the model achieving an accuracy of 75.8%. The analysis demonstrated that animal sex/reproductive status, age, and intake type are the strongest predictors of whether an animal will be adopted, providing valuable insights for shelter management decision-making.

These findings contribute to the growing body of research on animal welfare and shelter management, providing data-driven insights to support evidence-based practices in animal sheltering and adoption.

References

- ASPCA. 2025. "Shelter Intake and Adoption Statistics." <https://www.asPCA.org/helping-people-pets/shelter-intake-and-surrender/pet-statistics>.
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